The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 Apr 93.

METRIC

MIL-S-19500/587

14 January 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER
TYPES 1N6661, 1N6662, 1N6663, 1N6661US, 1N6662US, AND 1N6663US
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon diodes. Three levels of product assurance are provided for each device type as specified in MIL-S-19500 (see 6.2).

1.2 Physical dimensions. See figures 1 and 2.

1.3 Maximum ratings.

Type	V_{RWM}	I_O 4/ T_A = +150°C	1/ 2/ 3/ 4/ I_O T_A = +25°C	I_{FSM} t_p = 1/120s T_A = +25°C	T_{op} (ambient temperature)	T_{STG} (ambient temperature)	Barometric pressure reduced
	V (pk)	mA	mA	A	°C	°C	mmHg
1N6661, 1N6661US	225	150	500	5	-65 to +175	-65 to +200	8
1N6662, 1N6662US	400	150	500	5	-65 to +175	-65 to +200	8
1N6663, 1N6663US	600	150	500	5	-65 to +175	-65 to +200	8

1/ T_{EC} = 110°C.
2/ Devices with leads. Derate 2.8 mA/°C between 25°C and 150°C. Derate 6 mA/°C between 150°C and 175°C.
3/ US suffix devices. Derate 6.3 mA/°C at T_{EC} between 110°C and 150°C. Derate 6 mA/°C at T_{EC} between 150°C and 175°C.
4/ No forced air cooling or heat sinking shall be permitted for I_O ratings.

1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise indicated.

Type	V_F at I_F = 400 mA dc, 2% duty cycle, 8.3 ms max pulse width	I_R at T_A = +25°C		I_R at T_A = +150°C	
	V dc (max)	µA dc (max) at V_R		µA dc (max) at V_R	
1N6661, 1N6661US	1.0	.050	225 V dc	15	225 V dc
1N6662, 1N6662US	1.0	.050	400 V dc	15	400 V dc
1N6663, 1N6663US	1.0	.050	600 V dc	25	600 V dc

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

T_{EC} . . . . . Temperature of the end caps (mounting pads for US suffix devices).

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1 and 2 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Device construction. Devices supplied to this specification shall employ metallurgically bonded, thermally matched, non-cavity double plug construction. The metallurgical bond between the chip and the plugs shall meet category I requirements (see MIL-S-19500, appendix A for definitions).

3.4 Marking. Devices shall be marked as specified in MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the device, but shall be retained on the initial container.

3.4.1 Marking of U-suffix devices. All marking except for cathode identification, may be omitted from the body part.

3.4.2 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end. No color coding will be permitted. For U-suffix devices the cathode end may be identified by three evenly spaced dots of a contrasting color or a solid band.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be in accordance with MIL-S-19500 and table II herein.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3a	Temperature cycling per MIL-S-19500.	Temperature cycling per MIL-S-19500.
3c 1/	Thermal impedance (see 4.5.5)	Thermal impedance (see 4.5.5)
9	Table III, test 1 and 2	Not applicable
11	Table III, test 1, 2, 6, and 7	Table III, test 1 and 2
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; table III, test 6 and 7 and reverse scope display (see 4.5.3).	Subgroup 2 of table I herein; table III, test 6 and 7 and reverse scope display (see 4.5.3)

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.

4.3.1 Power burn-in conditions (method 1038, condition B (a) of MIL-STD-750). All devices shall be operated under one of the following conditions (see 4.5.1).

Type	T_A = +125°C; f = 60 Hz	T_A = +25°C; f = 60 Hz	T_A = +25°C
1N6661	V_R = 225 V (pk); I_O = 200 mA	V_R = 225 V (pk); I_O = 400 mA	I_F (dc) = 400 mA
1N6662	V_R = 400 V (pk); I_O = 200 mA	V_R = 400 V (pk); I_O = 400 mA	I_F (dc) = 400 mA
1N6663	V_R = 600 V (pk); I_O = 200 mA	V_R = 600 V (pk); I_O = 400 mA	I_F (dc) = 400 mA

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4.3.1.1 Alternate Mounting Conditions (For -US devcies). At the option of the manufacturer, other clips or heat sink mounting configurations may be utilized provided that one of the following conditions be met:

a. T_{EC} = +75°C to +125°C, V_{RWM} = rated, f = 50-60 Hz, I_O = rated I_{O2} (see 1.3.2).

b. Temporary attachment of leads or equivalent (thermal properties not to exceed the leaded part) T_j shall not exceed $T_{OP(max)}$ 50-60 Hz.

c. $I_O = \frac{I_{O2}}{2}$ to $2I_{O2(rated)}$, T_j = +125°C ±25°C, V_{RWM} = rated, 50-60 Hz.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table III herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

a. Subgroup 4: Conditions for intermittent operation life (LTPD) are as follows:

I_O = 400 mA; f = 60 Hz, T_A = +25°C ±3°C; t_{ON} = t_{OFF} = 3 minutes (minimum) for 2000 cycles; mounting conditions per MIL-STD-750, test method 1026 (see 4.5.1).

1N6661 = V_{RM} = 225 V(pk)
1N6662 = V_{RM} = 400 V(pk)
1N6663 = V_{RM} = 600 V(pk)

b. Subgroup 5: Conditions for accelerated steady-state operation life (LTPD) are as follows:

I_O = 400 mA dc; f = 60 Hz; T_A = +100°C; t = 96 hours (see 4.5.1).

1N6661 = V_{RM} = 225 V(pk)
1N6662 = V_{RM} = 400 V(pk)
1N6663 = V_{RM} = 600 V(pk)

c. Subgroup 6: Condition for thermal resistance is l = 3/8", $R_{\theta JL}$ = 160°C/W maximum (see 4.5.4).

4.4.2.2 Group B inspection, table IVb (JAN, JANTX and JANTXV) of MIL-S-19500.

a. Subgroup 2: Conditions for surge current are as follows:

MIL-STD-750, method 4066, T_A = +25°C, I_{FSM} = 5 A, ten 8.3 ms surges at 1 surge per minute. Mounting conditions per MIL-STD-750, method 1026.

b. Subgroup 3: Conditions for steady-state operation life (LTPD) are as follows:

I_O = 150 mA dc; f = 60 Hz; T_A = +150°C (see 4.5.1).

1N6661 = V_{RM} = 225 V(pk)
1N6662 = V_{RM} = 400 V(pk)
1N6663 = V_{RM} = 600 V(pk)

c. Subgroup 6: Condition for high-temperature life (nonoperating) (LTPD) is T_A = +200°C.

(Note: Leaded samples from the same lot may be used in lieu of U suffix samples for all subgroups.)

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Condition for terminal strength are as follows:

Lead tension: Test condition A; weight = 2 lbs; t = 30 ±3s.
Lead fatigue: Condition E.

NOTE: Not applicable to U-suffix devices.

b. Subgroup 6: Condition for steady-state operation life are as follows:

T_A = +150°C; f = 60 Hz; I_O = 150 mA (see 4.5.1).

1N6661 = V_{RM} = 225 V(pk)
1N6662 = V_{RM} = 400 V(pk)
1N6663 = V_{RM} = 600 V(pk)

NOTE: Leaded devices from the same lot may be used in lieu of U-suffix devices.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 State-state operation life. A half-sine wave of the specified peak voltage shall be impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conducted angle of the rectified current shall not be greater than 180° nor less than 150°.

4.5.2 Pulse measurements. Conditions of pulsed measurements shall be specified in section 4 of MIL-STD-750.

4.5.3 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 20 μA per division and 50 or 100 V per division as applicable. Reverse current over the knee shall be at least 100 μA and less than 200 μA. Any discontinuity or dynamic instability of the trace shall be cause for rejection of that device.

4.5.4 Thermal resistance. Thermal resistance shall be measured in accordance with method 3101 or 4081 of MIL-STD-750. On lead mounted devices the reference point shall be the lead temperature at .375 inches from the body of the device, and shall be held between 25°C and 85°C. For U-suffix devices, the reference point shall be the end cap, and shall be maintained at a temperature between 25°C and 110°C.

4.5.5 Thermal impedance. Thermal impedance $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit for $Z_{\theta JX}$ is screening (Table II of MIL-S-19500) shall be derived by each vendor by means of Process Control (PC). When three (minimum) successive sealing lots have exhibited control, the data from these three lots minimum will be used to establish a fixed screening limit, (not to exceed the group A Sub 2 Limit). Once a fixed limit has been established, monitor all future sealing lots using a three piece sample from each production lot to be plotted on the applicable X, R chart.

4.5.5.1 Thermal impedance conditions for group A ($Z_{\theta JX}$).

I_H = 10 A; t_H = 10 ms; I_M = 1 mA; t_{MD} = 70 μs.

Type	$Z_{\theta JX(max)}$
1N6661	20°C/W
1N6662	20°C/W
1N6663	20°C/W

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.5.5	$Z_{\theta JX}$		20	°C/W
Forward voltage	4011	I_F = 400 mA dc; 2% duty cycle max, 8.3 ms max pulse width; (pulsed; see 4.5.2)	V_{F1}	0.8	1.0	V dc
Reverse current	4016	D_C method	I_{R1}			
1N6661, 1N6661US		V_R = 225 V dc			0.050	µA dc
1N6662, 1N6662US		V_R = 400 V dc			0.050	µA dc
1N6663, 1N6663US		V_R = 600 V dc			0.050	µA dc
Breakdown voltage	4021	I_R = 100 µA dc	V_{BR1}			
1N6661, 1N6661US				270		V dc
1N6662, 1N6662US				480		V dc
1N6663, 1N6663US				720		V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Forward voltage	4011	I_F = 400 mA dc; 2% duty cycle max, 8.3 ms max pulse width; (pulsed; see 4.5.2)	V_{F2}	0.70	0.95	V dc
Reverse current	4016	D_C method	I_{R2}			
1N6661, 1N6661US		V_R = 225 V dc			25	µA dc
1N6662, 1N6662US		V_R = 400 V dc			25	µA dc
1N6663, 1N6663US		V_R = 600 V dc			25	µA dc
Low temperature operation:		T_A = -55°C				
Forward voltage	4011	I_F = 400 mA dc; 2% duty cycle max, 8.3 ms max pulse width; (pulsed; see 4.5.2)	V_{F3}		1.2	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 4						
Capacitance	4001	V_R = 4 V dc; f = 1 MHz; V_{sig} = 50 mVp-p maximum	C		20	pF
Subgroup 5						
Not applicable						
Subgroup 6 2/						
Surge current	4066	Mounting conditions per MIL-STD-750, test method 1026; T_A = 25°C; I_{FSM} = 5 A; ten 8.3 ms surges; 1 surge per minute				
Electrical measurements		See table III, steps 1, 2, and 5				

1/ For sampling plan, see MIL-S-19500.

2/ All devices required by the specified LTPD shall be randomly selected from the devices subjected to subgroup 2, 3, and 4.

TABLE II. Group E inspection (all quality levels) for qualification only.

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Inspection 1/	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Temperature shock (glass strain)	1056	500 cycles	
Hermetic seal	1071		
Electrical measurement		See table III, steps 1 and 2.	
Subgroup 2			22 devices c = 0
Burn-in (blocking life)	1038	V_R = 80% of rated V_{RWM} (see 1.3) T_A = 150°C; t = 1,000 hours min.	
Electrical measurement		See table III, steps 1, 3, and 5.	
Subgroup 3			3 devices c = 0
Not applicable			
Subgroup 4 2/			22 devices c = 0
Thermal resistance, junction to lead	3101 or 4081	L = 3/8"; $R_{\theta JL}$ = 160°C/W max. (see 4.5.5)	
Subgroup 5			15 devices c = 0 each type
Barometric pressure, reduced (altitude operation) Measurement	1001	8 mmHg; t = 60 s	
Measurement during test:			
Reverse current	4016	DC method	
1N6661, 1N6661US		V_R = 225 V dc; I_R = 0.050 µA dc maximum	
1N6662, 1N6662US		V_R = 400 V dc; I_R = 0.050 µA dc maximum	
1N6663, 1N6663US		V_R = 600 V dc; I_R = 0.100 µA dc maximum	

1/ For sampling plan, see MIL-S-19500.

2/ For U-suffix types thermal resistance, junction to end caps, $R_{\theta JEC}$ = 35°C/W max.

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TABLE III. Groups A, B, C, and E electrical measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Forward voltage	4011	I_F = 400 mA dc (pulsed); 2% duty cycle max; 8.3 ms max pulse width	V_{F1}	0.8 1/	1.0	V dc
2	Reverse current	4016	DC method	I_{R1}			
	1N6661, 1N6661US		V_R = 225 V dc			0.050	µA dc
	1N6662, 1N6662US		V_R = 400 V dc			0.050	µA dc
	1N6663, 1N6663US		V_R = 600 V dc			0.050	µA dc
3	Reverse current	4016	DC method	I_{R3}			
	1N6661, 1N6661US		V_R = 225 V dc			0.100	µA dc
	1N6662, 1N6662US		V_R = 400 V dc			0.100	µA dc
	1N6663, 1N6663US		V_R = 600 V dc			0.100	µA dc
4	Breakdown voltage	4021	I_R = 100 µA dc	V_{BR1}			
	1N6661, 1N6661US				270		V dc
	1N6662, 1N6662US				480		V dc
	1N6663, 1N6663US				720		V dc
5	Reverse current	4016	DC method	ΔI_{R1} 4/	100% of initial value or ± .010 µA dc, whichever is greater.		
	1N6661, 1N6661US		V_R = 225 V dc				
	1N6662, 1N6662US		V_R = 400 V dc				
	1N6663, 1N6663US		V_R = 600 V dc				
6	Forward current	4011	I_F = 400 mA dc (pulsed) 2% duty cycle max; 8.3 ms max pulse width	ΔI_{R2} 4/	±25 mV dc change from previously measured value.		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table III herein, steps 1, 2, and 4.
b. Subgroups 4 and 5, see table III herein, steps 1, 2, 4, 5, and 6.

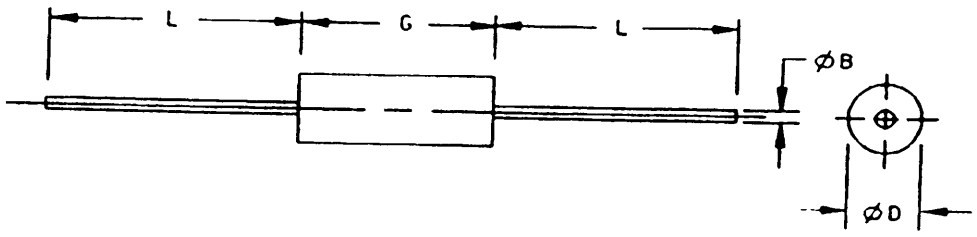
2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, and 4.
b. Subgroups 3 and 6, see table III herein, steps 1, 3, 4, 5, and 6.

3/ The electrical measurements for table V of MIL-S-19500 are as follows: Subgroups 2 and 3, see table III herein, steps 1, 2, and 4 for all levels.

4/ Devices which exceed the group A limits for this test shall not be accepted.

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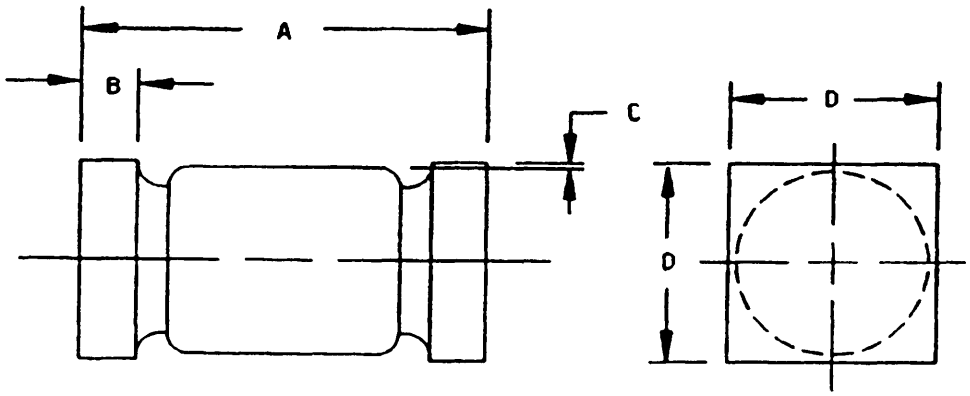


Symbol	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
G	3.05	5.08	.120	.200	
ØD	1.52	2.29	.060	.090	2
L	25.40	38.10	1.000	1.500	
ØB	0.46	0.56	.018	.022	3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The specified lead diameters apply in the zone between .050 inch (1.27 mm) from the diode body to the end of the lead. Outside of this zone the lead diameter shall not exceed the maximum of dimension ØD.

FIGURE 1. Physical dimensions (types 1N6661, 1N6662, and 1N6663).

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Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	4.19	4.95	.168	.200
B	0.48	0.71	.019	.028
C	0.08		.003	
D	1.78	2.16	.091	.103

NOTES:
1. Dimensions are in inches.
2. Metric equivalent are given for general information only.

FIGURE 2. Physical dimensions of surface mount family (types 1N6661US, 1N6662US, 1N6663US).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Type and quality assurance level.

6.3 Supersession. The 1N6661, 1N6662, and 1N6663 (MIL-S-19500/587) are preferred types for high reliability, high stress applications in lieu of 1N645, 1N647, and 1N649 (MIL-S-19500/240).

6.4 Metric conversion. This item was designed using inch-pound units of measurement. In case of problems involving conflicts between the metric and inch-pound units, the inch-pound units shall rule.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI
Navy - SH
Air Force - 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1405)

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